

MODULE-1

Introduction to Solid Waste Management Municipal Solid Waste Characteristics and quantities generation rates and waste composition; Integrated waste management issues, collection, recovery, reuse, recycling, energy-from-waste, and landfilling.

Solid waste refers to any discarded or unwanted materials. It includes various items such as paper, plastics, glass and food waste. The solid waste can include sludge from industrial plants or other discarded materials that result from industrial, commercial, mining and agricultural operations.

Solid waste can come from a variety of sources and consist of multiple types of trash generated from different human activities. These waste types can be categorized into several groups based on their origin, composition and characteristics:

- **Municipal Solid Waste (MSW):** This type includes everyday items discarded by households, such as paper, plastics, food scraps, textiles, packaging materials, yard waste and household hazardous waste like batteries or cleaning products.
- **Industrial Waste:** Generated by industrial processes and manufacturing activities, industrial waste comprises materials like scrap metal, chemicals, solvents, sludge and other by-products from factories and production facilities.
- **Commercial Waste:** Originating from commercial establishments such as offices, restaurants, shops and institutions, this waste type includes paper, packaging materials, food waste and other discarded items.
- **Construction and Demolition Debris (C&D):** Generated from construction, renovation and demolition activities, C&D waste includes concrete, wood, bricks, asphalt, metals, drywall and other materials used in building structures.
- **Hazardous Waste:** Materials that pose a risk to human health or the environment due to their toxic, flammable, corrosive or reactive nature are considered hazardous. This category includes items like batteries, fluorescent bulbs, pesticides, certain chemicals and medical waste.
- **Electronic Waste (E-waste):** Discarded electronic devices and equipment, such as computers, televisions, smartphones and appliances, can contain hazardous materials like lead, mercury and other components that require specialized handling and disposal.
- **Agricultural Waste:** Generated from farming activities, agricultural waste includes crop residues, animal manure, pesticides and other waste produced in agricultural processes.

Municipal Solid Waste (MSW)—more commonly known as trash or garbage—consists of everyday items we use and then throw away, such as product packaging, grass clippings, furniture, clothing, bottles, food scraps, newspapers, appliances, paint, and batteries. This comes from our homes, schools, hospitals, and businesses.

Sources of MSW

- **Homes:** Items like food scraps, clothing, furniture, and packaging
- **Businesses:** Items like packaging, office supplies, and used electronics
- **Schools:** Items like paper, food, and packaging

- **Hospitals:** Items like medical waste, which can be infectious or non-infectious

Types of MSW

- **Paper:** Newspapers, paper products, and other paper items
- **Plastics:** Plastic bottles, plastic plates, and plastic wrap
- **Glass:** Bottles and other glass items
- **Food:** Food scraps and other food items
- **Yard waste:** Grass clippings and other yard waste
- **Textiles:** Clothing, leather, and other textiles
- **Metals:** Cans and other metal items

Source of Municipal Solid Waste	Type of Solid Waste
Residential	Food waste, food container and packer, can, bottles, papers and newspaper, clothes, garden waste, e-wastes, furniture waste
Commercial Centre (office lot, small shop, restaurant)	Vary type of papers and boxes, food waste, food container and packer, can, bottles,
Institutional (school, university, college, hospital)	Office waste, food waste, garden waste, furniture waste
Industry (factory)	Office waste, cafeteria waste, processing waste
City Centre (drainage and road)	Vary type of garden waste, construction waste, public waste

Source: Franklin Association (1999)

WASTE GENERATION AND COMPOSITION

Information on waste quantity and composition is important in evaluating alternatives in terms of equipment, systems, plans, and management programs. For example, if wastes generated at a commercial facility consist of only paper products, the appropriate equipment is shredders and balers. Similarly, on the basis of the quantity generated, we can plan appropriate means for separation, collection, and recycling programs. That is to say, the success of SWM depends on the appropriate assessment of the quantity of waste generated.

Waste generation

Waste generation encompasses those activities in which waste, be it solid or semi-solid material, no longer has sufficient economic value for its possessor to retain it. The processing of raw materials is the first stage when wastes are generated, and waste generation continues thereafter at every step in the process as raw materials are converted into final products for consumption. Figure 1 below shows a simplified material-flow diagram indicating the path of generation of solid wastes.

Generation

It is suggested that we can reduce the amount of solid waste by limiting the consumption of raw materials and increasing the rate of recovery and reuse. There needs to be, therefore, a societal change in the perception of wastes. This sounds simple But, implementing changes in the society is difficult,

unless appropriate management solutions are provided. That said, we also must note that the changes in waste generation contribute to changes in waste composition. Figure 1 shows the material flow and the generation of waste in the society.

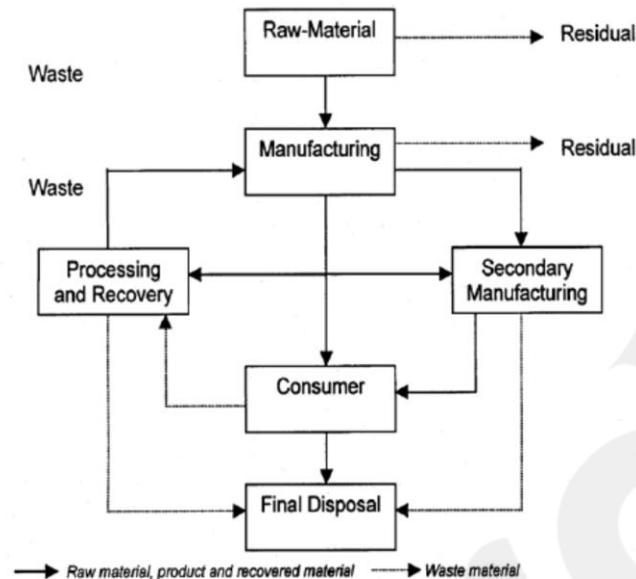


Fig 1: Material Flow and Waste

Waste composition:

Some of the general observations associated with the composition of wastes include the following: The major constituents are paper and decomposable organic materials. More often than not, metal, glass, ceramics, textile, dirt and wood form part of the composition, and their relative proportion depends on local factors. Average proportions of the constituents reaching the disposal sites are consistent and urban wastes are fairly constant although subject to long-term changes such as seasonal variations. Waste composition varies with the socio-economic status within a particular community, since income, for example, determines life style, composition pattern and cultural behaviour. Figure 2 and 3 illustrates this phenomenon in India:

Type of waste	Total solid waste generation daily (kg)	Solid waste composition (%)	Average solid waste generation (kg/persons/day)
Food waste	102.40	21.60	0.06
Plastic	184.60	39.40	0.11
Glass	39.70	8.40	0.02
Paper	75.50	15.90	0.04
Metal	43.60	9.20	0.02
Other	22.20	4.70	0.01
Total	468.00	100.00	0.27

Fig 2: Generation and Composition of waste

Waste component	Low - income group (%)	Middle-income group (%)	High-income group (%)	Average waste generated/week (%)
Paper	14	19	25	20
Cans	12	13	5	11
Glass	20	10	7	11
Plastics	14	17	24	18
Food wastes	25	36	37	34
Garden waste	15	5	2	6

Fig 3: Total waste composition from the three income groups

Waste composition also depends on the moisture content, density and relative distribution of municipal wastes, as shown in Fig 4 below, and is important for the characterisation of solid waste for most applications.

Component	Mass by %	Moisture content (%)	Density kg/m3
Food waste	20	70	290
Paper	30	6	85
Plastics	10	2	65
Textile	5	10	65
Leather	5	10	160
Wood	10	20	240
Glass	10	2	195
Garden trimmings	10	60	105

Fig 4: Waste composition on the moisture content, density and by mass

Note that the density of waste changes as it moves from the source of generation to the point of ultimate disposal, and such factors as storage methods, salvaging activities, exposure to weather, handling methods and decomposition influence the density. In short, predicting changes of waste composition is as difficult as forecasting waste quantities.

Factors causing variation

- As we know, wastes cause pollution. While the nature of wastes determines the type and intensity of pollution, it also helps us decide on the appropriate application, engineering design and technology for management. For example, the nature of wastes has implications for collection, transport and recycling. For effective SWM, therefore, we not only need information about the present but
- also the expected future quantity and composition of wastes. There are several factors, which affect the present as well as the future waste quantity and composition, and some of which are listed below:
- **Geographic location:** The influence of geographic location is related primarily to different climates that can influence both the amount of certain types of solid wastes generated and the collection operation. For instance, substantial variations in the amount of yard and garden wastes generated in Various parts of India are related to the climate. To illustrate, in the warmer southern areas, where the growing season is considerably longer compared to the northern areas, yard wastes are collected in considerably larger quantities and over a longer period of time.
- **Seasons:** Seasons of the year have implications for the quantities and composition of certain types of solid wastes. For example, the growing season of vegetables and fruits affect the quantities of food wastes.
- **Collection frequency:** A general observation is that in localities, where there are ultimate collection services, more wastes are *collected*. Note that this does not mean that more wastes are *generated*. For example, if a homeowner has access to only one or two containers per week, due to limited container capacity, he or she will store newspapers or other materials in some specified storage area. However, the same homeowner will tend to throw them away, if there is access to unlimited container services. In this latter situation, the quantity of waste generated may actually be the same but the quantity collected, as it relates to the frequency of collection, is considerably different.
- **Population diversity:** The characteristics of the population influence the quantity and composition of waste generated. The amount of waste generated is more in low-income areas compared to that in high-income areas. Similarly, the composition differs in terms of paper and other recyclables, which are typically more in high-income areas as against low income areas (see Table 2.1).
- **Extent of salvaging and recycling:** The existence of salvaging and recycling operation within a community definitely affects the quantity of wastes collected.

- **Public attitude:** Significant reduction in the quantity of solid waste is possible, if and when people are willing to change – on their own volition – their habits and lifestyles to conserve the natural resources and to reduce the economic burden associated with the management of solid wastes.
- **Legislation:** This refers to the existence of local and state regulations concerning the use and disposal of specific materials and is an important factor that influences the composition and generation of certain types of wastes. The Indian legislation dealing with packing and beverage container materials is an example. In short elements that relate to waste generation include land use characteristics, population in age distribution, legislation, socio economic conditions, household and approximate number.

WASTE CHARACTERISTICS

In order to identify the exact characteristics of municipal wastes, it is necessary that we analyse them using physical and chemical parameters .

Physical characteristics

Information and data on the physical characteristics of solid wastes are important for the selection and operation of equipment and for the analysis and design of disposal facilities. The required information and data include the following:

- **Density:** Density of waste, i.e., its mass per unit volume (kg/m^3), is a critical factor in the design of a SWM system, e.g., the design of sanitary landfills, storage, types of collection and transport vehicles, etc. To explain, an efficient operation of a landfill demands compaction of wastes to optimum density. Any normal compaction equipment can achieve reduction in volume of wastes by 75%, which increases an initial density of 100 kg/m^3 to 400 kg/m^3 . In other words, a waste collection vehicle can haul four times the weight of waste in its compacted state than when it is uncompacted. A high initial density of waste precludes the achievement of a high compaction ratio and the compaction ratio achieved is no greater than 1.5:1. Significant changes in density occur spontaneously as the waste moves from source to disposal, due to scavenging, handling, wetting and drying by the weather, vibration in the collection vehicle and decomposition.

Note that:

- *the effect of increasing the moisture content of the waste is detrimental in the sense that dry density decreases at higher moisture levels;*
- *soil-cover plays an important role in containing the waste;*
- *there is an upper limit to the density, and the conservative estimate of in-place density for waste in a sanitary landfill is about 600 kg/m^3 .*

- **Moisture content:** Moisture content is defined as the ratio of the weight of water (wet weight - dry weight) to the total weight of the wet waste. Moisture increases the weight of solid wastes, and thereby, the cost of collection and transport. In addition, moisture content is a critical determinant in the economic feasibility of waste treatment by incineration, because wet waste consumes energy for evaporation of water and in raising the temperature of water vapour. In the main, wastes should be insulated from rainfall or other extraneous water. We can calculate the moisture percentage, using the formula given below: A typical range of moisture content is

20 to 40%, representing the extremes of wastes in an arid climate and in the wet season of a region of high precipitation. However, values greater than 40% are not uncommon.

$$\text{Moisture content (\%)} = \frac{\text{Wet weight} - \text{Dry weight}}{\text{Wet weight}} \times 100$$

- **Size:** Measurement of size distribution of particles in waste stream is important because of its significance in the design of mechanical separators and shredders. Generally, the results of size distribution analysis are expressed in the manner used for soil particle analysis. That is to say, they are expressed as a plot of particle size (mm) against percentage, less than a given value.
- **Compressibility of MSW:** Degree of physical changes of the suspended solids or filter cake when subjected to pressure.

Chemical characteristics

Knowledge of the classification of chemical compounds and their characteristics is essential for the proper understanding of the behaviour of waste, as it moves through the waste management system. The products of decomposition and heating values are two examples of chemical characteristics. If solid wastes are to be used as fuel, or are used for any other purpose, we must know their chemical characteristics, including the following:

- **Lipids:** This class of compounds includes fats, oils and grease, and the principal sources of lipids are garbage, cooking oils and fats. Lipids have high heating values, about 38,000 kJ/kg (kilojoules per kilogram), which makes waste with high lipid content suitable for energy recovery. Since lipids become liquid at temperatures slightly above ambient, they add to the liquid content during waste decomposition. Though they are biodegradable, the rate of biodegradation is relatively slow because lipids have a low solubility in water.
- **Carbohydrates:** These are found primarily in food and yard wastes, which encompass sugar and polymer of sugars (e.g., starch, cellulose, etc.) with general formula $(CH_2O)_x$. Carbohydrates are readily biodegraded to products such as carbon dioxide, water and methane. Decomposing carbohydrates attract flies and rats, and therefore, should not be left exposed for long duration.
- **Proteins:** These are compounds containing carbon, hydrogen, oxygen and nitrogen, and consist of an organic acid with a substituted amine group (NH_2) . They are mainly found in food and garden wastes. The partial decomposition of these compounds can result in the production of amines that have unpleasant odours.
- **Natural fibres:** These are found in paper products, food and yard wastes and include the natural compounds, cellulose and lignin, that are resistant to biodegradation. (Note that paper is almost 100% cellulose, cotton over 95% and wood products over 40%.) Because they are a highly combustible solid waste, having a high proportion of paper and wood products, they are suitable for incineration. Calorific values of oven-dried paper products are about half that for fuel oil, which is 44,200 kJ/kg.
- **Synthetic organic material (Plastics):** Accounting for 1 – 10%, plastics have become a significant component of solid waste in recent years. They are highly resistant to biodegradation and, therefore, are objectionable and of special concern in SWM. Hence the increasing attention being paid to the recycling of plastics to reduce the proportion of this waste component at disposal sites. Plastics have a high heating value, about 32,000 kJ/kg, which makes them very

suitable for incineration. But, you must note that polyvinyl chloride (PVC), when burnt, produces dioxin and acid gas. The latter increases corrosion in the combustion system and is responsible for acid rain.

- **Non-combustibles:** This class includes glass, ceramics, metals, dust and ashes, and accounts for 12 – 25% of dry solids.
- **Heating value:** An evaluation of the potential of waste material for use as fuel for incineration requires a determination of its heating value, expressed as kilojoules per kilogram (kJ/kg). The heating value is determined experimentally using the *Bomb calorimeter test*, in which the heat generated, at a constant temperature of 25 C from the combustion of a dry sample is measured. Since the test temperature is below the boiling point of water (100 C), the combustion water remains in the liquid state. However, during combustion, the temperature of the combustion gases reaches above 100 C, and the resultant water is in the vapour form.

Note that while evaluating incineration as a means of disposal or energy recovery, we need to consider the heating values of respective constituents as shown in figure 5.

For example:

- Organic material yields energy only when dry.
- The moisture content in the waste reduces the dry organic material per kilogram of waste and requires a significant amount of energy for drying.
- The ash content of the waste reduces the proportion of dry organic material per kilogram of waste and retains some heat when removed from the furnace.

Material used as fuel	Gross Calorific value, GCV (MJ/kg)
Coal	23.0 – 32.0
Fuel oil	40.0 – 45.0
Natural gas	50.0 – 55.0 (38.6 MJ/m ³)
Plastic	27.0 – 34.0
Municipal solid waste	8.5 – 11.0
Hospital & clinical waste	17.5 – 22.5
Chemical waste	18.5 – 23.0
Sewage sludge	7.0 – 13.0 (depending on dryness)
Vehicle tyres	32.0 – 40.0
Sugar cane bagasse	8.0 – 12.5 (depending on dryness)
Wood	17.0 – 20.0
Rice husks, rice straw	12.0 – 18.0
Straw	14.0 – 15.5
Meat & bone meal	20.0 – 28.0 (depending on fat content)
Poultry litter	13.0 – 14.0

Fig 5: Typical calorific values for the waste materials

Ultimate analysis: This refers to an analysis of waste to determine the proportion of carbon, hydrogen, oxygen, nitrogen and sulphur, and the analysis is done to make mass balance calculation for a chemical or thermal process. Besides, it is necessary to determine ash fraction because of its potentially harmful environmental effects, brought about by the presence of toxic metals such as cadmium, chromium, mercury, nickel, lead, tin and zinc. Note that other metals (e.g., iron, magnesium, etc.) may also be present but they are non-toxic. Figure 6 shows the classification of composition of proximate and ultimate analysis and of a typical values municipal solid waste composition.

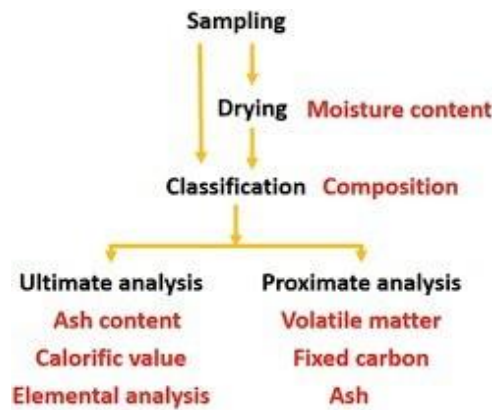


Fig 6: Classification of Proximate and Ultimate analysis of MSW

Proximate analysis: This is important in evaluating the combustion properties of wastes or a waste or refuse-derived fuel. The fractions of interest are:

- moisture content, which adds weight to the waste without increasing its heating value, and the evaporation of water reduces the heat released from the fuel;
- ash, which adds weight without generating any heat during combustion;
- volatile matter, i.e., that portion of the waste that is converted to gases before and during combustion;
- fixed carbon, which represents the carbon remaining on the surface grates as charcoal. A waste or fuel with a high proportion of fixed carbon requires a longer retention time on the furnace grates to achieve complete combustion than a waste or fuel with a low proportion of fixed carbon.

Proximate Analysis		Ultimate Analysis	
Moisture	16.9 (wt%)	C (carbon)	53.84 (wt%)
Volatile	55.1 (wt%)	H (hydrogen)	5.73 (wt%)
Fixed carbon	8 (wt%)	O (oxygen)	32.93 (wt%)
Ash	20 (%)	N (nitrogen)	1.68 (wt%)
Calorific value	2388 (kcal/kg)	S (sulfur)	0.87 (wt%)

Fig 7: Typical values for Proximate and Ultimate analysis of MSW

To evaluate alternative processing and recovery options (e.g., incineration process), we need information on the chemical characteristics of wastes, and wastes can typically be a combination of Combustible and non-combustible materials.

Waste mismanagement, e.g., the practice of throwing wastes into streets, storm water drains, vacant land, etc., leads to breeding of disease vectors such as rats, with their attendant fleas carrying germs, etc., which results in an epidemic such as plague, malaria, etc. This has an adverse impact on public health and the environment. We will look into the consequences of waste mismanagement on Public health and the environment.

What is solid waste management?

Solid Waste Management refers to the collection, transportation, processing, recycling, treatment, and disposal of solid waste in a way that reduces its environmental and health impacts. The goal is to handle waste efficiently while promoting sustainability, resource recovery, and public health.

Elements of Solid waste Management: Solid Waste Management (SWM) consists of several key elements that ensure waste is handled efficiently from generation to disposal. These elements include is shown in Figure 8,

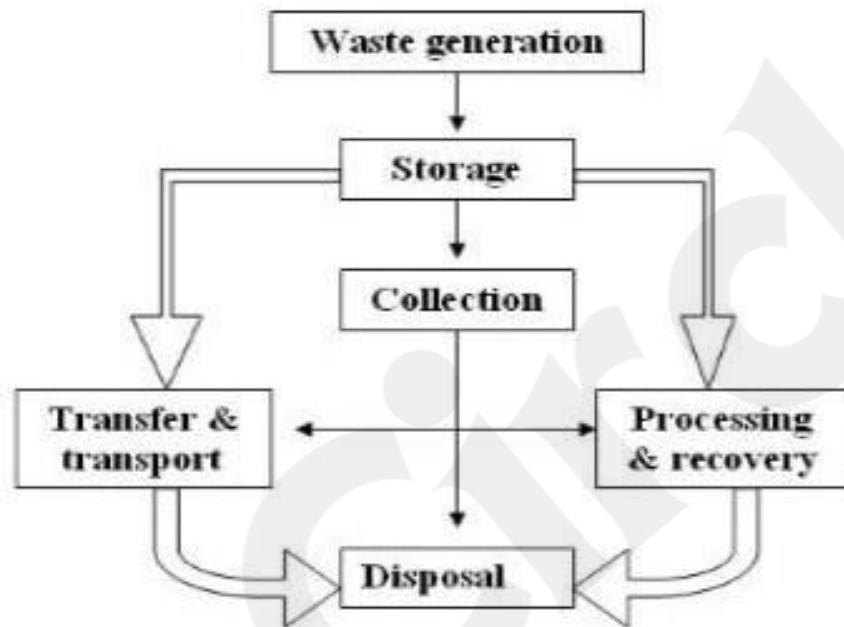


Fig 8: Flowchart showing the Elements of Solid Waste Management.

1. Waste Generation

- The initial stage where waste is produced by households, industries, businesses, hospitals, agriculture, and construction activities.
- Waste types include organic, recyclable, hazardous, and construction waste.

2. Waste Storage & Segregation

- **Storage:** Temporary containment of waste in bins or containers before collection.
- **Segregation:** Sorting waste at the source into categories such as biodegradable, recyclable, hazardous, and non-recyclable to facilitate proper disposal and recycling.

3. Waste Collection

- Waste is collected from households, commercial areas, and industrial sites using collection vehicles.

- Collection methods include door-to-door collection, communal bins, and curbside pickup.
4. Waste Transportation
- Waste is transported to transfer stations, recycling plants, or disposal sites using trucks, compactors, and waste transport vehicles.
 - Efficient transportation reduces costs and environmental impact.
5. Waste Processing & Recovery
- Recycling:** Conversion of materials like paper, glass, metal, and plastic into new products.
 - Composting:** Organic waste is decomposed to produce manure for agriculture.
 - Incineration:** Burning waste at high temperatures to generate energy.
 - Waste-to-Energy (WTE):** Converting waste into electricity or biofuels.
 - Anaerobic Digestion:** Breaking down organic waste in oxygen-free conditions to produce biogas.
6. Waste Disposal
- Sanitary Landfills:** Properly designed waste disposal sites with protective liners and leachate management systems.
 - Open Dumping (Not Recommended):** Uncontrolled waste disposal that causes pollution.
 - Hazardous Waste Disposal:** Specialized treatment for toxic and biomedical waste.

INTEGRATED WASTE MANAGEMENT (IWM)

Integrated Waste Management (IWM) is a systematic approach to handling municipal solid waste (MSW) that combines various waste management practices (reduction, recycling, composting, energy recovery, and disposal) to achieve sustainability and efficiency.

The elements of Integrated Waste Management is shown in Figure 9 . However, several challenges and issues affect the implementation of IWM worldwide.

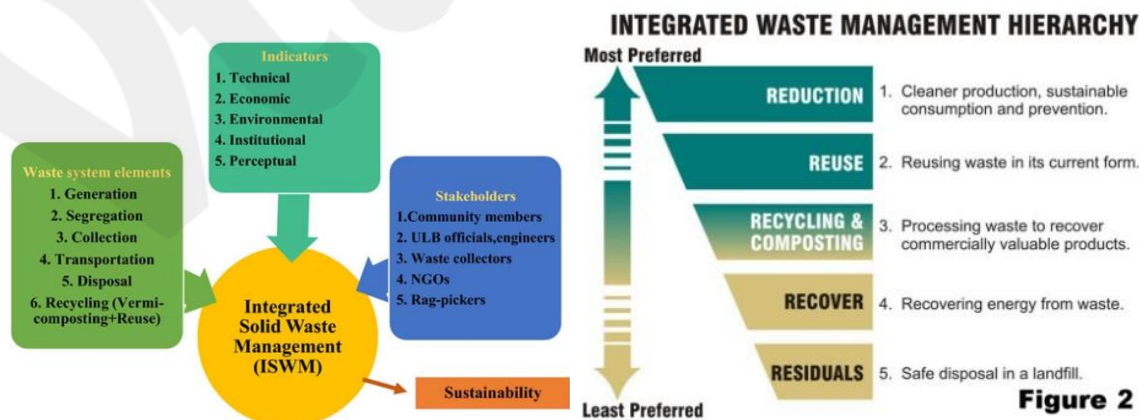


Figure 2

Fig 9: Elements of IWM

How IWM works

- **Waste reduction:** Reduce the amount of waste you create by buying less and reusing items
- **Waste recycling:** Recycle paper, plastic, and other materials
- **Waste composting:** Compost organic materials to improve soil quality
- **Waste treatment:** Use bio-gasification, thermal treatment, and other methods to treat waste
- **Waste disposal:** Landfill waste as a last resort, or use waste to create energy

Benefits of IWM

- **Environmental:** Reduces pollution and the amount of waste in landfills
- **Economic:** Reduces costs for waste disposal
- **Social:** Improves social acceptance of waste management

Integrated Waste Management issues:

Integrated Waste Management (IWM) include: inadequate collection infrastructure, poor waste segregation, low recycling rates, improper disposal practices, lack of public awareness, high implementation costs, limited treatment options, and potential environmental impacts from waste-to-energy facilities; all of which can lead to environmental pollution, health hazards, and inefficient resource utilization if not properly addressed.

Breakdown of the issues:

- **Collection and Segregation:**
 - Insufficient collection points: Not enough bins or designated areas for proper waste separation, leading to mixed waste streams.
 - Inconsistent collection schedules: Unreliable collection services can cause waste accumulation and improper disposal.
 - **Lack of public awareness on segregation:** People might not understand how to properly sort waste, resulting in contamination.
- **Treatment and Disposal:**
 - Overreliance on landfills: Improperly managed landfills can generate harmful methane gas and leachate, polluting groundwater.
 - Limited recycling capacity: Insufficient recycling facilities and processing capabilities can hinder recycling efforts.
 - Waste-to-energy concerns: Concerns about air pollution and potential toxic emissions from waste incineration plants.
- **Economic and Social Challenges:**
 - High implementation costs: Setting up a comprehensive IWM system requires significant investment in infrastructure, technology, and public awareness campaigns.
 - Lack of political will: Insufficient government support and commitment to enforce waste management regulations.
 - Community resistance: Potential opposition to waste treatment facilities due to concerns about environmental impact and aesthetics.
- **Environmental Impacts:**
 - Air pollution: Open burning of waste and emissions from poorly managed landfills contribute to air pollution.
 - Water contamination: Leachate from landfills can pollute groundwater sources.
 - Land degradation: Improper disposal practices can lead to soil contamination and land degradation.

Potential Solutions:

- **Public awareness campaigns:**

Educating citizens on waste reduction, reuse, and proper waste segregation.

- **Improved collection infrastructure:**

Increasing the number of waste collection bins and implementing efficient collection routes.

- **Stricter regulations and enforcement:**

Implementing policies to incentivize waste reduction and recycling, and penalize improper disposal.

- **Investing in recycling technology:**

Enhancing recycling facilities and processing capabilities to improve recycling rates.

- **Community engagement:**

Involving local communities in decision-making regarding waste management strategies.

Differences Between Solid Waste Management (SWM) and Integrated Solid Waste Management (ISWM) is shown in Table 1.

Table 1: comparison of Solid Waste Management (SWM) and Integrated Waste Management (IWM):

Aspect	Solid Waste Management (SWM)	Integrated Waste Management (IWM)
Definition	The process of collecting, transporting, treating, and disposing of solid waste.	A comprehensive approach that integrates multiple strategies (reduce, reuse, recycle, recover, and dispose) for sustainable waste management.
Approach	Focuses mainly on waste collection and disposal.	Uses a combination of strategies such as waste prevention, recycling, composting, and energy recovery before disposal.
Waste Hierarchy	Primarily focused on disposal methods like landfills and incineration.	Follows the waste management hierarchy: Reduce → Reuse → Recycle → Recover → Dispose.
Sustainability	Less sustainable if reliant on landfilling and incineration.	Promotes long-term sustainability by minimizing waste generation and maximizing resource recovery.
Environmental Impact	May contribute to pollution if waste is not properly managed.	Reduces environmental impact through eco-friendly practices.
Economic Efficiency	Can be costly due to landfilling and waste transportation.	Aims to reduce costs by encouraging resource recovery and recycling.
Community Involvement	Generally managed by municipalities with limited community participation.	Encourages community involvement, education, and policies that promote waste reduction.

Aspect	Solid Waste Management (SWM)	Integrated Waste Management (IWM)
Regulatory Framework	Governed by waste collection and disposal laws.	Requires an integrated policy framework combining regulations, economic incentives, and public participation.

Key Difference:

- SWM is a **traditional** approach mainly concerned with **waste disposal**.
- IWM is a **modern, sustainable approach** that prioritizes **waste reduction, recycling, and resource recovery** before disposal.

COMPREHENSIVE WASTE MANAGEMENT PROGRAMS:

A Comprehensive Waste Management Program is a structured system that integrates multiple waste management strategies to minimize environmental impact, promote sustainability, and ensure efficient waste handling. These programs involve waste reduction, collection, recycling, treatment, and disposal while encouraging public participation and policy enforcement. Figure 10 shows the typical working of waste management programs in a city.



Fig 10: Waste Management programs in a City.

COLLECTION:

Municipal Solid Waste (MSW) collection refers to the process of gathering, transporting, and disposing of household, commercial, and institutional waste generated within a municipality. Effective MSW collection is crucial for maintaining public health, environmental cleanliness, and sustainability. The collection component in MSW includes,

1. Collection Services: Collection Services includes not only the gathering of solid wastes, but also the hauling of wastes after collection to the location where the collection vehicle is emptied. Collection of solid wastes is one of the most costly functional elements, because of high cost of fuel and labour.

Various types of collection services now used for municipal, commercial, and industrial sources are:

- **Curb Service:** The house owner is responsible for placing the solid waste containers at the curb on the scheduled day. The workers come, collect, and empty the container and put back at the curb.
- **Alley Service:** The containers placed at the alley line from where they are picked up by workers from refuse vehicle who deposit back the empty container.
- **Set out set back service:** Set out man go to the houses to collect containers and empty them in the refuse vehicles. Another group of persons returns them to the house owner's yard.
- **Backyard Service:** The workers with the vehicles carry a bin, wheel-borrow to the yard and empty the solid waste container in it.

2. Collection frequency: Climatic conditions and requirements of a locality as well as containers and costs determine the collection frequency. The quality of solid waste containers on site also determines the collection frequency. While deciding collection frequency, therefore, you must consider the following:

- cost, e.g., optimal collection frequency reduces the cost as it involves fewer trucks, employees and reduction in total route distance;
- storage space, e.g., less frequent collection may require more storage
- space in the locality;
- Sanitation, e.g., frequent collection reduces concerns about health, safety and nuisance associated with stored refuse.

3. Storage containers: Proper container selection can save collection energy, increase the speed of collection and reduce crew size. Containers should also be durable, easy to handle, and economical, as well as resistant to corrosion, weather and animals. The containers may fall under either of the following two categories:

(i) Stationary containers: These are used for contents to be transferred to collection vehicles at the site of storage.

(ii) Hauled containers: These are used for contents to be directly transferred to a processing plant, transfer station or disposal site for emptying before being returned to the storage site.

4. Collection crew: A "collection crew" in solid waste management refers to a team of workers responsible for physically collecting household and commercial waste from designated collection points (like bins or dumpsters) and transporting it to a transfer station or landfill.

The optimum crew size for a community depends on labour and equipment costs, collection methods and route characteristics. The size of the collection crew also depends on the size and type of collection vehicle used, space between the houses, waste generation rate and collection frequency.

5. Collection route: An efficient routing of collection vehicles helps decrease costs by reducing the labour expended for collection. Proper planning of collection route also helps conserve energy and minimise working hours and vehicle fuel consumption.

Routing (network) analyses and planning can increase the likelihood of all streets being serviced equally and consistently; help supervisors locate or track crews quickly; provide optimal routes that can be tested against driver judgement and experience.

6. Transfer station: A transfer station, or resource recovery centre, is a building or processing site for the temporary deposition, consolidation and aggregation of waste.

It is an intermediate station between final disposal option and collection point in order to increase the efficiency of the system, as collection vehicles and crew remain closer to routes. In some instances, the transfer station serves as a pre- processing point, where wastes are dewatered, scooped or compressed.

Challenges in MSW Collection

- Inefficient route planning and high operational costs.
- Lack of public awareness and participation in waste segregation.
- Limited infrastructure and improper disposal leading to pollution.
- Inadequate workforce and outdated equipment.

RECOVERY:

Recovery of waste means any operation the principal result of which is waste serving a useful purpose by replacing other materials which would otherwise have been used to fulfil a particular function, or waste being prepared to fulfil that function, in the plant or in the wider economy.

It refers to processes that extract useful materials or energy from waste, reducing landfill dependency and promoting sustainability. Waste recovery is a key element in modern solid waste management strategies, supporting a circular economy. The waste can be recovered by two facilities: Materials Recovery Facilities (MRFs) and Refuse Derived Fuel (RDF). They are both solid waste management systems that process waste to recover materials and generate energy.

1. Materials Recovery Facilities (MRFs): MRF is a specialized plant that receives, sorts, and prepares recyclable materials like paper, plastic, glass, and metal from waste, separating them for further processing and sale to end-user manufacturers, essentially maximizing the amount of reusable materials recovered from waste.

Recovery from MRF:

- **Paper & Cardboard:** Repulping for new paper products.
- **Plastics:** Reprocessing into new plastic products or alternative fuels.
- **Metals:** Melting and reusing in manufacturing.
- **Glass:** Crushing and remelting for new glass products.

2. Refuse Derived Fuel (RDF): RDF is a fuel made from leftover materials after recyclable materials have been removed from waste. Waste is shredded, treated, compressed into RDF, and then baled.

RDF is burned in Energy from Waste (EfW) plants to generate electricity. RDF is renewable, easy to store and transport, and has low emissions, thus diverting waste from landfills and recovering value from it.

RDF-Energy Recovery (Waste-to-Energy - WtE)

- **Incineration:** Burning waste to generate heat or electricity.
- **Gasification & Pyrolysis:** Converting waste into synthetic gas, fuels, or chemicals.
- **Landfill Gas Recovery:** Capturing methane from decomposing waste to generate energy.

REUSE:

The most preferred option for waste management in the hierarchy is to prevent the generation of waste at various stages including in the design, production, packaging, use, and reuse of products. Reuse is the practice of using a material again instead of throwing it away. It's a waste management strategy that helps reduce energy and resource consumption.

Examples of Reuse

- **Donate:** Give away items you no longer use to friends, family, charities, or secondhand stores
- **Repurpose:** Find new uses for old items, like using old jars for storage or old towels for wash rags
- **Reuse containers:** Use empty food containers for leftovers, or reuse plastic bags for trash or pet waste
- **Swap:** Trade items with friends or coworkers
- **Bring your own bags:** Use reusable bags instead of single-use plastic bags.

Benefits of Reuse

- Reduces waste and the amount of material that goes to landfills
- Saves money by avoiding the cost of buying new items
- Helps others in your community

RECYCLING:

Recycling is the process of turning waste materials into new products. It's a key waste management strategy that helps reduce pollution, conserve resources, and save energy.

Recycling involves collecting waste materials, processing them into raw materials, and then using those raw materials to make new products. Many materials can be recycled, including glass, paper, aluminum, cardboard, plastic, wood, textiles, and electronic equipment.

Figure 11 shows that recycling is a key component of sustainable waste management and the circular economy.



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Fig 11: Circular Economy with Recycling

Stages of the recycling process:

- **Collection:** Gathering used materials from households, businesses, and designated recycling bins.
- **Sorting:** Separating different types of recyclable materials like paper, plastic, glass, metal, etc., often done manually or through automated sorting technologies.
- **Cleaning:** Washing and removing contaminants from the collected materials to ensure quality of the recycled product.
- **Processing:** Depending on the material, this might involve shredding, melting, or breaking down the waste into smaller pieces suitable for reprocessing.
- **Manufacturing new products:** Using the recycled material as a raw input to create new products like plastic bottles, paper, aluminum cans, etc.

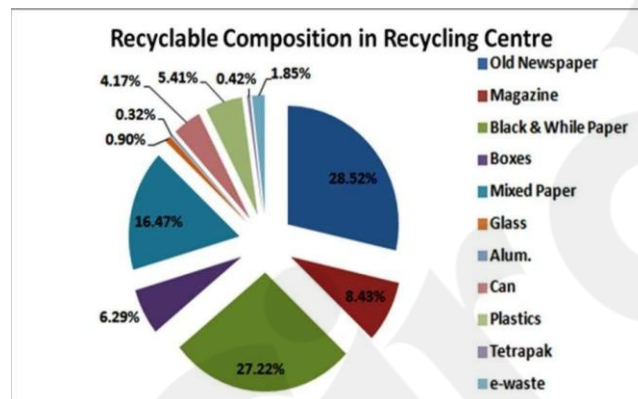


Fig 12: Composition of Recyclable materials

Benefits of recycling and Figure 12 shows the composition that can be recycled.

- **Reduces waste:** Recycling diverts waste from landfills, which are a major source of pollution.
- **Conserves resources:** Recycling reduces the need to harvest new materials.
- **Saves energy:** Recycling uses less energy than producing new materials.
- **Reduces pollution:** Recycling reduces greenhouse gas emissions and air and water pollution.

ENERGY-FROM-WASTE:

Where material recovery from waste is not possible, energy recovery from waste through production of heat, electricity, or fuel is preferred. Bio-methanation, waste incineration, production of refuse derived fuel (RDF), co-processing of combustible non biodegradable dry fraction from MSW in cement kilns and pyrolysis or gasification are some waste-to-energy technologies. **Energy-from-Waste (EfW)** is the process of converting non- recyclable waste materials into usable energy, such as electricity, heat, or fuel. It helps reduce landfill waste while generating renewable energy, making it a key component of sustainable waste management. The Methods of Energy Recovery from Waste are as follows,

(i) **Incineration (Combustion)**

Incineration is a waste treatment process that burns waste in a furnace at high temperatures (900–1200°C), broken down and oxidized and thus heat and electricity are recovered and Figure 13 shows the Incineration Plant.

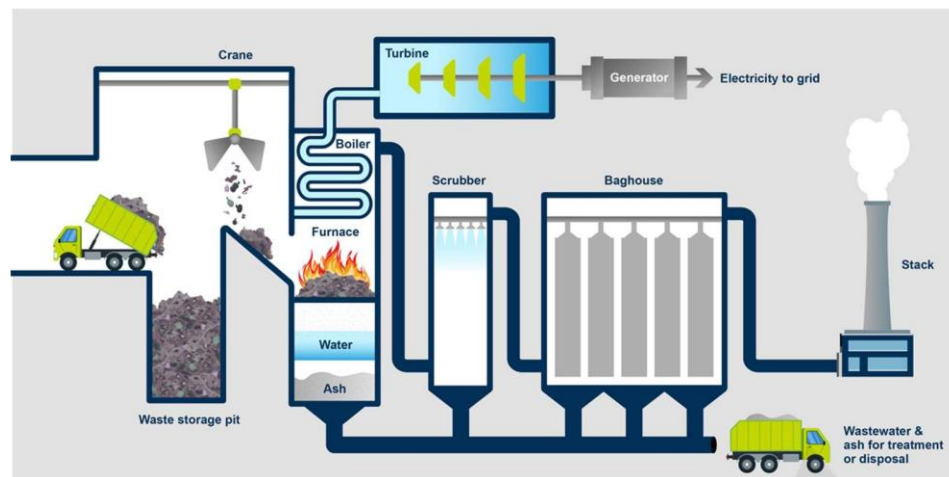


Fig 13: Incineration Plant

Benefits and drawbacks

- Incineration can reduce waste volume by up to 90%
- Incineration can help extend the lifespan of landfills
- However, incineration is expensive to build and operate
- Incineration can release pollutants into the air, water, and food supply
- Incineration can increase the risk of cancer, birth defects, and other health issues

(ii) **Gasification:** Gasification is a process that converts organic or fossil-based carbonaceous materials at high temperatures ($>700^{\circ}\text{C}$), without combustion, with a controlled amount of oxygen and/or steam into carbon monoxide, hydrogen, and carbon dioxide. The gasification process is shown in Figure 14.

- Waste is heated in a low-oxygen environment to produce **syngas**. (synthetic gas).
- Syngas is used to generate electricity or refined into fuels like hydrogen.

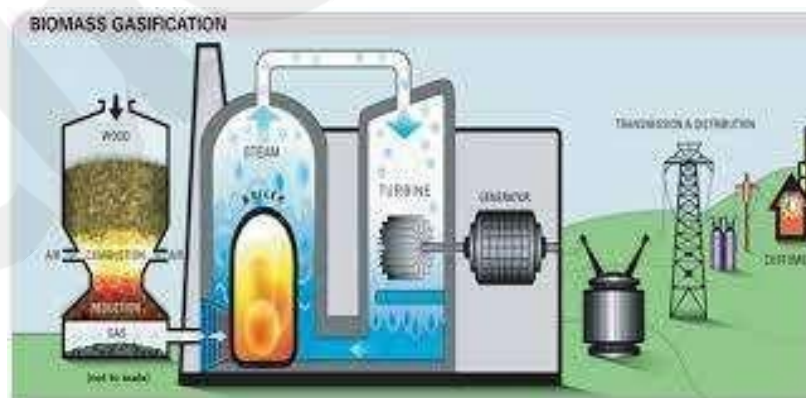


Fig 14: Gasification Plant

(iii) **Pyrolysis:** Pyrolysis is one of the technologies available to convert biomass to an intermediate liquid product that can be refined to drop-in hydrocarbon biofuels, oxygenated fuel additives and petrochemical replacements. Pyrolysis is the heating of an organic material, such as biomass, in the absence of oxygen as shown in Figure 15..

- Similar to gasification but at higher temperatures and without oxygen.
- Produces bio-oil, syngas, and biochar, which can be used for energy and soil improvement.

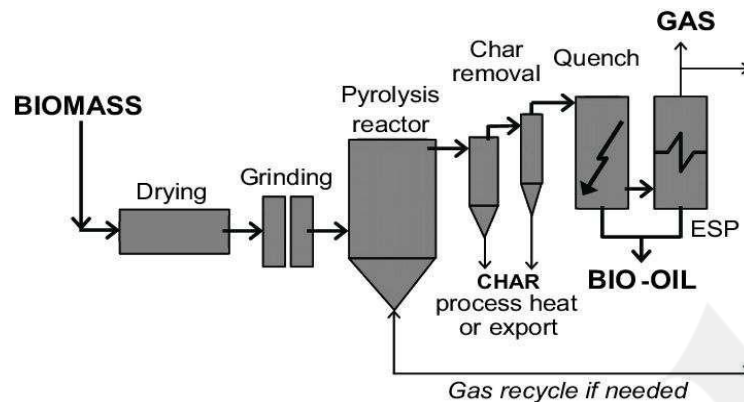


Fig 15: Pyrolysis unit

INCINERATION VS PYROLYSIS VS GASIFICATION

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INCINERATION	PYROLYSIS	GASIFICATION
Incineration is a waste treatment method that involves burning waste materials at high temperatures, often producing heat or electricity	Pyrolysis is a thermal treatment process where organic materials are heated without oxygen, causing them to decompose into gases, liquids, and solid char	Gasification is a process that converts organic materials into a combustible gas (syngas) by reacting them with a controlled amount of oxygen or steam at elevated temperatures
Operates at high temperatures with abundant oxygen, leading to complete combustion	Involves heating in the absence of oxygen, causing partial combustion and producing char and liquids	Uses a controlled amount of oxygen or steam to convert materials into a mixture of gases (syngas)
Typically results in ash and gases, with a focus on energy recovery	Results in char, bio-oil, and gases, offering the potential for bioenergy and biofuels	Results in syngas, which can be used for various applications, including energy production and chemical synthesis
Suitable for a wide range of waste types, including mixed and hazardous wastes	Well-suited for organic materials like biomass, plastics, and tires	Can handle a variety of feedstocks, including coal, biomass, and municipal solid waste

Fig 16: Difference between Incineration, Pyrolysis and Gasification

(iv) Anaerobic Digestion

- Organic waste (food scraps, manure, sewage sludge) is broken down by microorganisms in an oxygen-free environment.
- Produces **biogas** (methane + CO₂), which can be used for electricity, heating, or as a vehicle fuel.

(v) Landfill Gas Recovery

Landfill gas recovery is the process of capturing and using the gases produced in landfills. The recovered gas can be used to generate power or heat, or sold to natural gas pipelines.

The working of Landfill gas is shown in Figure 17.

- Wells and a vacuum system extract the gas from the landfill
- The gas is collected through pipes and directed to a central point
- The gas is processed and treated depending on its intended use

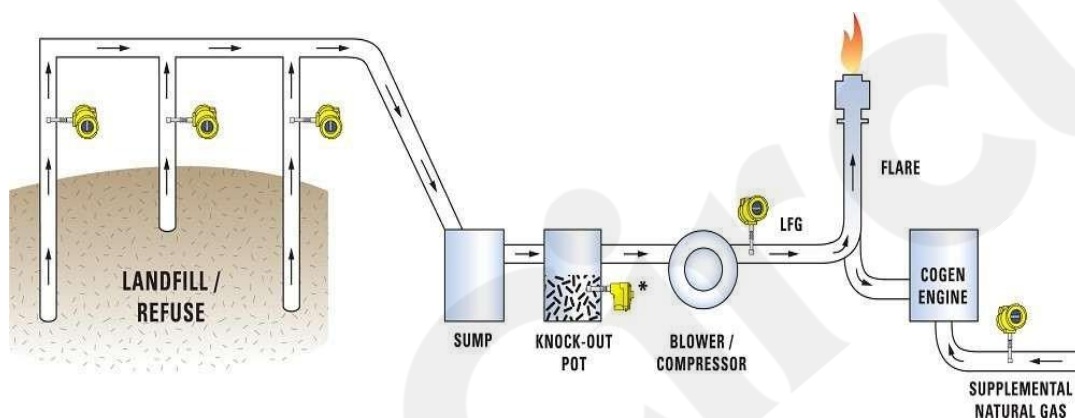


Fig 17: Landfill Gas Recovery.

Benefits

- **Reduced emissions:** Landfill gas recovery reduces greenhouse gas emissions and air pollutants
- **Improved air quality:** Landfill gas recovery helps reduce odors and other hazards associated with landfill gas emissions
- **Energy independence:** Landfill gas recovery can help improve energy independence
- **Cost savings:** Landfill gas recovery can help create cost savings

Regulations

- The MSW (Management and Handling) Rules, 2000 stipulates that landfill sites must have a gas collection system to minimize odor and prevent harmful gases from migrating off-site
- The rules also specify that methane gas emissions at landfill sites must not exceed 25% of the Lower Explosive Limit (LEL)

(vi) **Landfilling:** It is the process of disposing of municipal solid waste (MSW) by burying it in designated sites as shown in figure 18. While landfills are the most common method of waste disposal, modern landfills are engineered to minimize environmental and health impacts. Residual inert wastes at the end of the hierarchy are to be disposed in sanitary lined landfills, which are constructed in accordance with stipulations prescribed in SWM Rules, 2016. All over the world, landfills which integrate

the capture and use of methane are preferred over landfills which do not capture the landfill gas. As per the hierarchy, the least preferred option is the disposal of waste in open dumpsites. However, Indian laws and rules do not permit disposal of organic matter into sanitary landfills and mandate that only inert rejects (residual waste) from the processing facilities, inert street sweepings, etc. can be landfilled. In cases where old dumps are to be closed, there is a possibility of capturing methane gas for further use. However, repeated burning of waste significantly decreases the potential of capturing methane.



Fig 18: Landfill Waste

The types of landfill are as follows,

1. Sanitary Landfills

- Designed with **protective liners** (clay or synthetic) to prevent groundwater contamination.
- Waste is compacted and covered daily with soil or other materials.
- Includes gas collection and leachate treatment systems.

2. Controlled Landfills

- Less advanced than sanitary landfills but have some waste management controls.
- May have basic lining and leachate management.

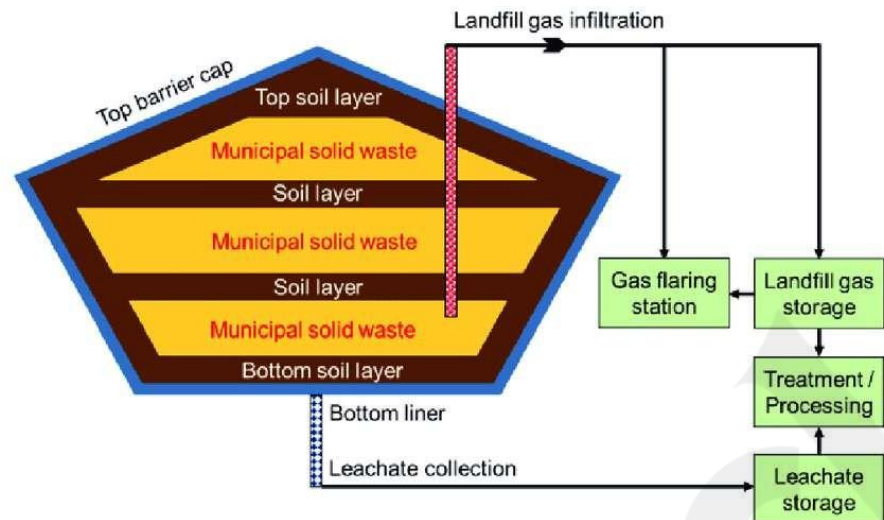
3. Open Dumps (Uncontrolled Landfills)

- Unregulated waste disposal sites.
- High risk of pollution, fires, disease spread, and environmental hazards.
- Being phased out in many countries.

4. Bioreactor Landfills

- Actively managed to **accelerate waste decomposition** by adding moisture and aeration.
- Generates more landfill gas, which can be captured for energy use

Landfilling Process: The daily operation at a municipal solid waste landfill includes the tipping of waste into a specific area of the landfill, called the working face, followed by compaction or crushing of the waste and covering it with soil at day's end and figure 19 shows post landfill process.



(a)



(b)

Fig 19: Process in Landfilling

Step 1: Site Selection & Preparation

- Location chosen **away from water bodies, residential areas, and flood-prone zones.**
- **Liners installed** (clay or synthetic) to prevent groundwater contamination.

Step 2: Waste Deposition

- Waste is brought in **trucks and compacted** to reduce volume.
- Daily covering with **soil, plastic, or other materials** to prevent odor and pests.

Step 3: Leachate Collection & Treatment

- **Leachate (liquid waste)** is collected using pipes and treated in treatment plants.
- Prevents **groundwater and soil contamination.**

Step 4: Gas Collection & Utilization

- Decomposing waste generates **methane (CH₄)** and **carbon dioxide (CO₂).**
- Modern landfills **capture landfill gas** for energy production (**waste-to-energy systems**).

Step 5: Final Closure & Land Rehabilitation

- Once full, the landfill is **sealed with clay, soil, and vegetation**.
- Land can be repurposed into **parks, solar farms, or green spaces**.

Environmental & Health Impacts of Landfills**(i) Environmental Impacts**

- **Leachate pollution** – Can contaminate groundwater and soil.
- **Methane emissions** – A potent greenhouse gas contributing to climate change.
- **Odor and air pollution** – Decomposing waste produces **hydrogen sulfide (H₂S)** and other gases.
- **Land degradation** – Permanent loss of usable land.
- **Wildlife & pest attraction** – Birds, rodents, and insects gather around open dumps.

(ii) Health Impacts

- **Respiratory issues** from toxic landfill gases.
- **Waterborne diseases** from contaminated groundwater.
- **Cancer risks** from long-term exposure to hazardous waste.
- **Skin and eye infections** for workers and nearby residents.